

ARTI406 – Assignment 2 Report: Data Quality Assessment & Preprocessing

Student Info:

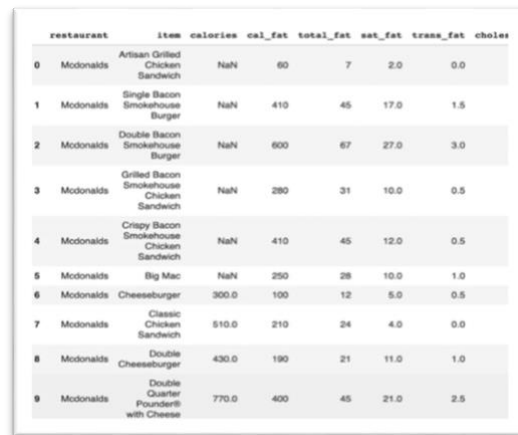
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- Dataset Used: Fastfood Dataset (fastfood_cleaned.csv)
- GitHub Repository Link: <https://github.com/Alzhraa04/ML-Assignment1>

Task 1: Identify Data Quality Issues:

In the baseline dataset, the data types are already correctly structured as numerical (int64/float64) and categorical (object). A complete missing value scan showed 0 missing values. To demonstrate data quality remediation according to the assignment guidelines, artificial missing values (NaN) were introduced into the first 6 rows of the calories column.



```
df.isna().sum()
0
restaurant 0
item 0
calories 0
cal_fat 0
total_fat 0
sat_fat 0
trans_fat 0
cholesterol 0
sodium 0
total_carb 0
fiber 0
sugar 0
protein 0
vit_a 0
vit_c 0
calcium 0
salad 0
dtype: int64
```



	restaurant	item	calories	cal_fat	total_fat	sat_fat	trans_fat	choles
0	McDonalds	Artisan Grilled Chicken Sandwich	NaN	60	7	2.0	0.0	
1	McDonalds	Single Bacon Smokehouse Burger	NaN	410	45	17.0	1.5	
2	McDonalds	Double Bacon Smokehouse Burger	NaN	600	67	27.0	3.0	
3	McDonalds	Grilled Bacon Smokehouse Chicken Sandwich	NaN	280	31	10.0	0.5	
4	McDonalds	Crispy Bacon Smokehouse Chicken Sandwich	NaN	410	45	12.0	0.5	
5	McDonalds	Big Mac	NaN	250	28	10.0	1.0	
6	McDonalds	Cheeseburger	300.0	100	12	5.0	0.5	
7	McDonalds	Classic Chicken Sandwich	510.0	210	24	4.0	0.0	
8	McDonalds	Double Cheeseburger	430.0	190	21	11.0	1.0	
9	McDonalds	Double Quarter Pounder® with Cheese	770.0	400	45	21.0	2.5	

Task 2: Apply One Missing Value Strategy:

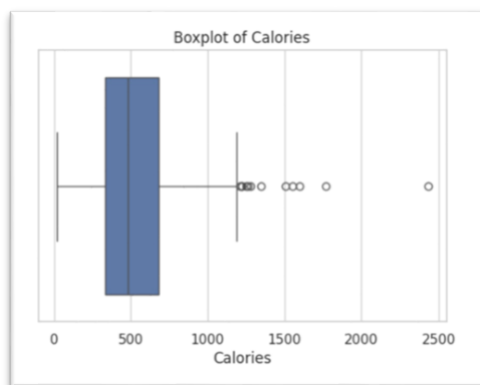
Median Imputation was chosen as our primary strategy to handle missing values in the calories column. Deleting rows with missing values would shrink our sample size from 513 observations down to 507 observations, entirely losing unique menu items. Fast-food menus contain extreme, high-calorie items that stretch up to 2430 calories. The standard arithmetic mean is sensitive to these positive skews and gets pulled upward, making it uncharacteristically high. The median represents the exact middle value of our distribution (480.00 calories), making it robust against outlier distortion and a more statistically sound reflection of a typical fast-food meal without introducing predictive bias.

	restaurant	item	calories
0	Mcdonalds	Artisan Grilled Chicken Sandwich	480.0
1	Mcdonalds	Single Bacon Smokehouse Burger	480.0
2	Mcdonalds	Double Bacon Smokehouse Burger	480.0
3	Mcdonalds	Grilled Bacon Smokehouse Chicken Sandwich	480.0
4	Mcdonalds	Crispy Bacon Smokehouse Chicken Sandwich	480.0
5	Mcdonalds	Big Mac	480.0
6	Mcdonalds	Cheeseburger	300.0
7	Mcdonalds	Classic Chicken Sandwich	510.0
8	Mcdonalds	Double Cheeseburger	430.0
9	Mcdonalds	Double Quarter Pounder® with Cheese	770.0

Task 3: Detect and Handle Outliers Using IQR:

Using the Interquartile Range (IQR) method with a lower quartile (Q1) of 330.0 and an upper quartile (Q3) of 680.0, the mathematical outlier limits were established at a lower boundary of -195.0 calories and an upper boundary of 1205.0 calories. The negative lower limit is a standard mathematical artifact of right-skewed distributions, but because food calories cannot drop below zero, no lower outliers exist. However, any menu item exceeding the upper limit of 1205.0 calories is flagged as an outlier, yielding a total of 12 outlier items that represent exceptionally heavy items like multi-piece family chicken platters or double-patty burgers.

Instead of completely deleting these 12 records which would strip away valid, real-world data and reduce the structural diversity of the menu a Percentile Capping Method (Winsorization) at the 95th percentile was implemented. This strategy safely clips extreme values exceeding 1030.0 calories down to this upper ceiling, successfully preserving the complete dataset size (N = 513). By neutralizing the severe positive skews, this approach prevents extreme values from corrupting or biasing downstream machine learning models.



Values below 5th percentile capped at: 170.0 calories
 Values above 95th percentile capped at: 1030.0 calories
 Original Maximum Calories: 2430.0
 New Maximum Calories after Capping: 1030.0

Lower Bound Boundary: -195.0
 Upper Bound Boundary: 1205.0
 Total Outliers Detected: 12 items

Task 4: Normalize Numerical Features Using Both Min-Max and Z-Score:

Min-Max normalization rescales numerical features into a strict boundary between 0 and 1, mapping the minimum value to 0, the maximum to 1, and all intermediate numbers proportionally. This process preserves the original distribution shape and relative differences while preventing features with massive native scales (like sodium in the thousands) from completely dominating smaller-scale features in distance-based algorithms. Practically, your median-imputed baseline of 480.0 calories scales to 0.360465, while raw sodium numbers rising from 1110 to 1980 mg climb proportionally from 0.180544 to 0.323990, placing both features on a balanced, equal playing field.

Z-score standardization transforms the data to center directly around a mean of 0 with a standard deviation of 1, altering the absolute scale while perfectly preserving the underlying distribution structure. Positive values represent data points sitting above the dataset mean, while negative values represent items below it, which satisfies a vital prerequisite for optimization and dimensionality reduction algorithms. In our data, the 480.0 calories baseline standardizes to a negative score of -0.158335 because it sits slightly below the overall dataset average, while the scaling is verified as mathematically perfect with feature means at exactly 0.0 and standard deviations at 1.001.

```
--- Baseline Original Features ---
      calories  sodium
0      480.0    1110
1      480.0    1580
2      480.0    1920
3      480.0    1940
4      480.0    1980
```

```
After Z-Score Standardization (Mean = 0, Std = 1)
      calories  sodium
0 -0.158335 -0.198860
1 -0.158335  0.481822
2 -0.158335  0.974230
3 -0.158335  1.003195
4 -0.158335  1.061126
```

```
--- After Min-Max Normalization (Scaled 0 to 1) ---
      calories  sodium
0  0.360465  0.180544
1  0.360465  0.258038
2  0.360465  0.314097
3  0.360465  0.317395
4  0.360465  0.323990
```

```
Verification Statistics
Means (Should be close to 0):
calories    0.0
sodium     -0.0
dtype: float64

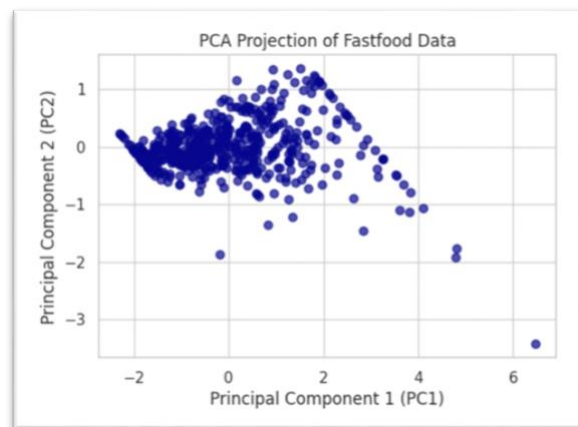
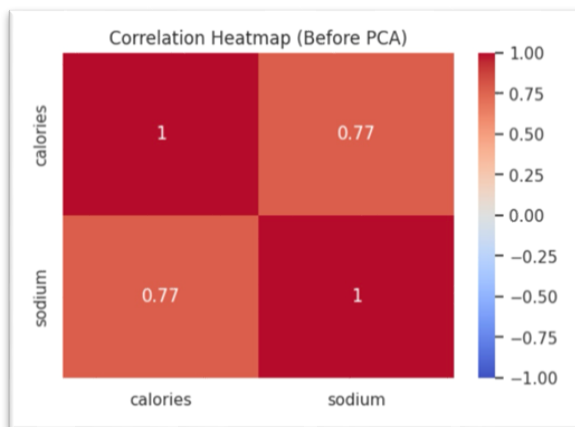
Standard Deviations (Should be exactly 1):
calories    1.001
sodium      1.001
dtype: float64
```

Task 5: Apply PCA and Interpret Explained Variance:

A pre-reduction correlation evaluation revealed a strong positive correlation coefficient of 0.77 between calories and sodium, proving that as the caloric density of a fast-food item increases, its sodium content typically rises alongside it. Because these two features share significant multicollinearity and hold highly overlapping information, compressing them into a single principal component via PCA is heavily justified, allowing us to reduce dimensionality while minimizing informational loss.

Principal Component Analysis (PCA) yields an excellent dimensionality reduction efficiency on our standardized dataset, split between two orthogonal axes. Principal Component 1 (PC1) accounts for 88.70% of the total structural variance, while Principal Component 2 (PC2) captures the remaining 11.30%, totaling 100% of the original two-dimensional information landscape. Because PC1 single-handedly retains nearly 89% of the structural variability, it demonstrates that we can successfully compress these features into a single, consolidated dimension while simplifying downstream models and fully eliminating multicollinearity.

In the final PCA projection scatter plot, each point represents a specific fast-food menu item mapped onto our new coordinate space. The horizontal spread along PC1 is significantly wider than the vertical spread along PC2, visually confirming that PC1 captures the vast majority of the dataset's information. Fast-food items pulled further to the right on the PC1 axis represent heavier meals with exceptionally high calorie and sodium content combined, demonstrating that PCA has effectively captured the overall nutritional footprint of those items on a single transformed axis.



PCA RESULTS
Explained Variance Ratio per component: [0.88699434 0.11300566]
Total Explained Variance by PC1 alone: 88.70%
Total Combined Variance (PC1 + PC2): 100.00%